PRESIGHT EUREKA 3.0 · DEFENSE & POLICING

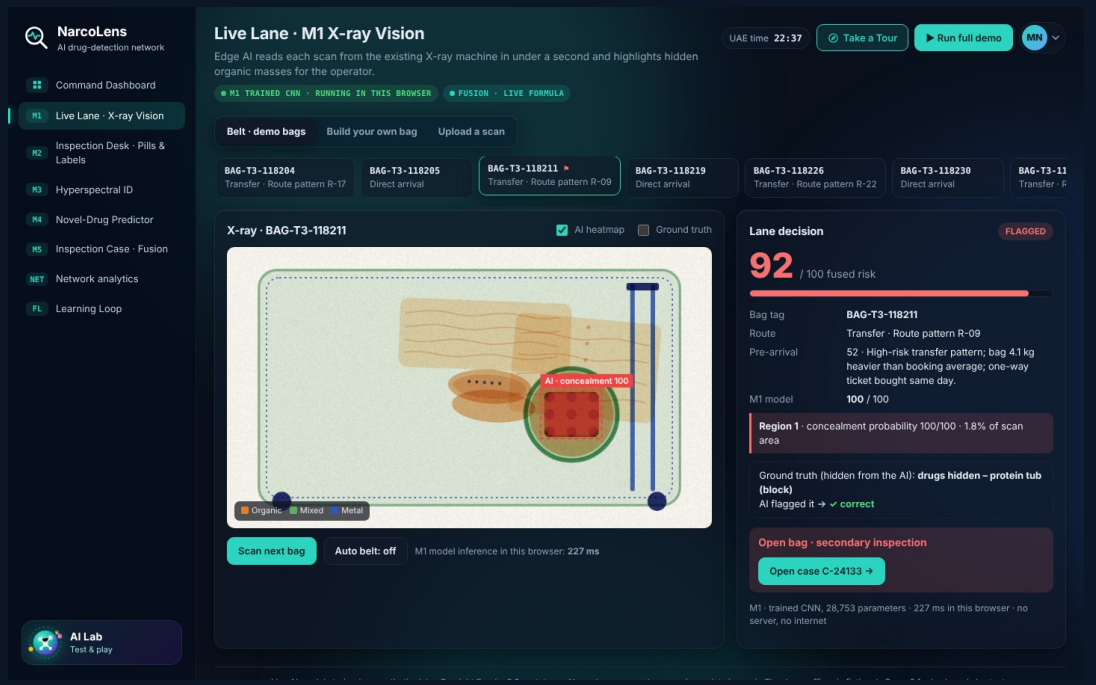
NarcoLens

AI-Assisted Substance Screening for Police and Airport Customs

Existing scanners | **Officer decides** | **Evidence-ready**

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Technology readiness: TRL 3, a proof of concept built and measured on synthetic data



Prototype: M1 flags a block hidden inside a protein tub (simulated scan)

Every suspicious substance demands a fast, defensible decision

502

seizures by Dubai Customs, January to May 2026

406 kg

of controlled substances intercepted

2.3 million

narcotic tablets intercepted

The current situation

The UAE already deploys advanced inspection systems, AI-driven risk analysis and K9 units. These figures show the volume of findings that officers must assess.

The remaining gap

After a flag, officers still rely on operator interpretation of X-ray images and on laboratory confirmation to identify the substance. Evidence is recorded across separate systems.

The operational consequence

Decisions made under this uncertainty can delay legitimate travellers and parcels, or leave the basis for a seizure harder to defend.

Source: Dubai Customs figures via Emirates News Agency (WAM), June 2026; also reported by Gulf News and Khaleej Times.

Four gaps remain after an item is flagged

Image interpretation

Concealments are found by operators reading dense X-ray images. In the public sources we reviewed, we found no published system that flags narcotics automatically in UAE customs scans.

Substance identity

Naming a substance depends on laboratory confirmation, so decisions at the inspection desk are made under uncertainty.

Emerging variants

New synthetic variants are designed to fall outside detection libraries, so recognition can trail the first seizure.

Evidence records

Findings, photos and decisions sit in separate systems, which makes the basis of a decision harder to reconstruct.

Why now: edge computers and hyperspectral cameras are increasingly available, scan data is already digital, and the UAE is investing in sovereign AI. What is missing is the software layer that connects them into one decision.

One intelligence layer across existing customs scanners

Delivered as software: an edge computer beside each scanner and a hyperspectral camera at the inspection desk

M1 · X-ray vision

Flags concealed organic masses and shows where the model found them.

M2 · Tablet and label checks

Verifies tablet imprints and aims to read Arabic and English labels against a registry.

M3 · Hyperspectral ID

Assesses the substance through its packaging and separates mixtures.

M4 · Variant predictor

Aims to add emerging synthetic variants to the library before their first seizure.

M5 · Explainable risk fusion

Combines the evidence into one score that shows each module's contribution.

Officer decision and evidence record

The officer decides. Every action is logged for laboratory referral.

Federated learning aims to let each site learn from laboratory-confirmed seizures without sharing raw images.

One bag, from first scan to laboratory referral

1

Before arrival

Route and booking patterns give a starting score. No nationality or ethnicity is used.

2

At the scanner

M1 flags a dense organic mass and highlights where.

3

Routing

M5 fuses the evidence and sends the bag to secondary inspection.

4

At the desk

M2 checks the label and tablet imprints from the officer's photo.

5

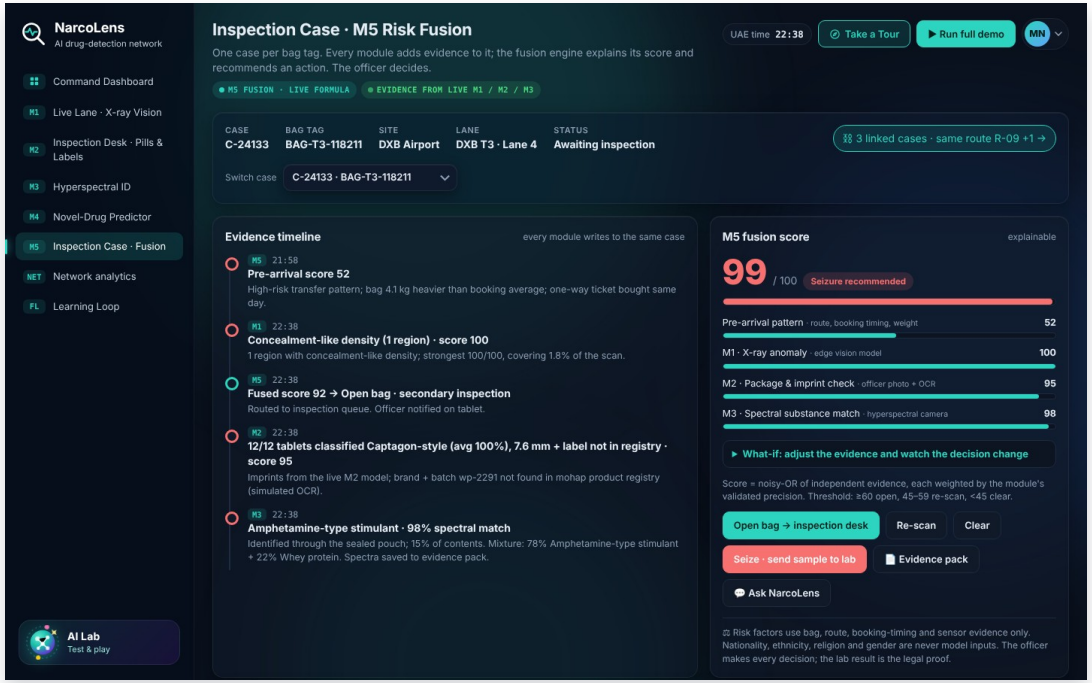
Hyperspectral scan

M3 assesses the substance through the sealed packaging.

6

Decision and referral

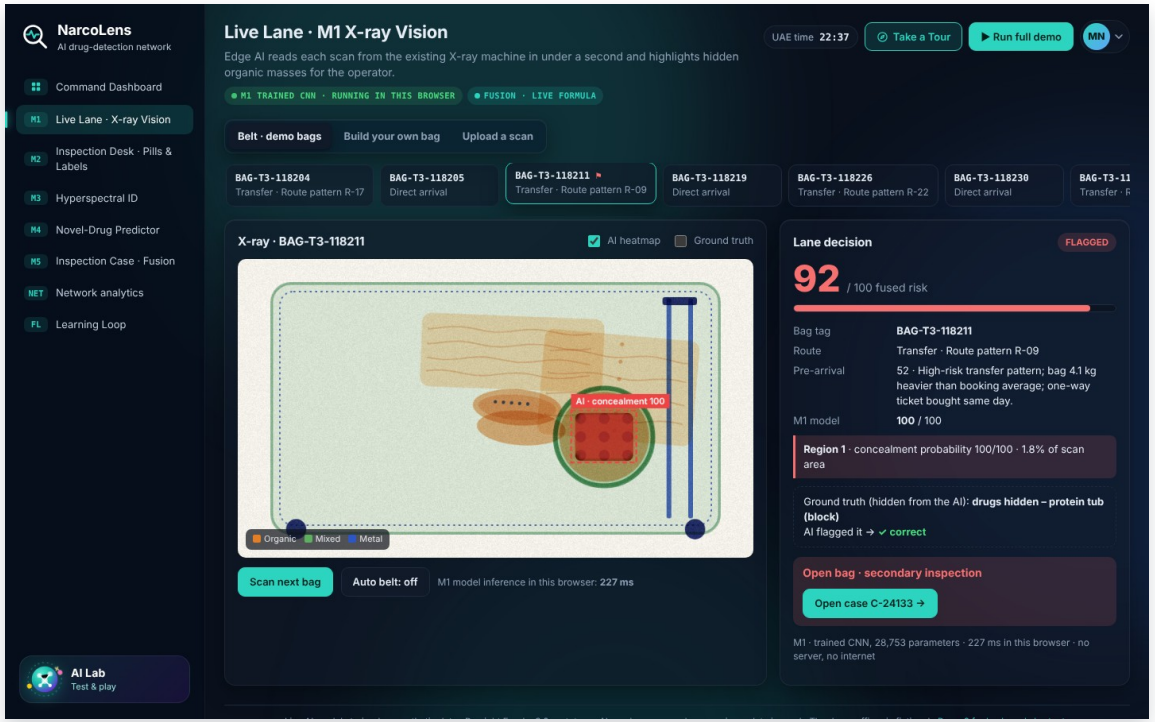
The officer decides. The case is recorded for laboratory referral.



Prototype: one case collects evidence from every module (simulated data)

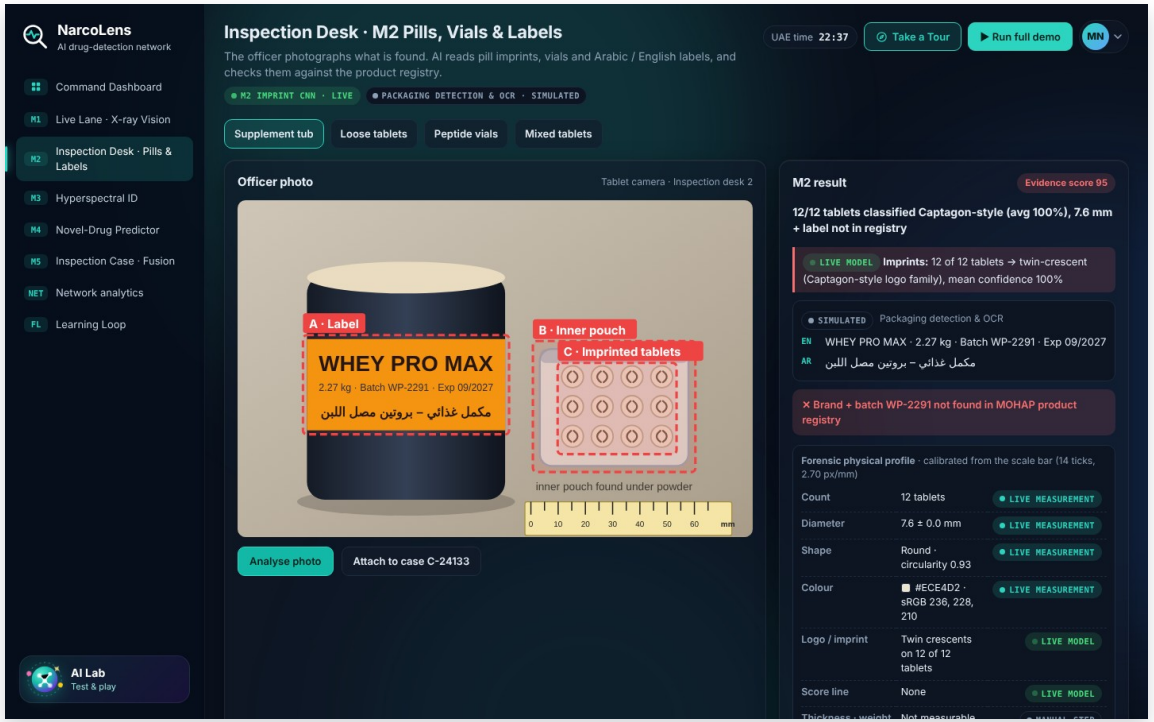
Cargo and postal parcels follow the same flow. The air waybill or parcel barcode becomes the case identifier.

Seeing what is hidden: X-ray and tablet checks



M1 · X-ray vision

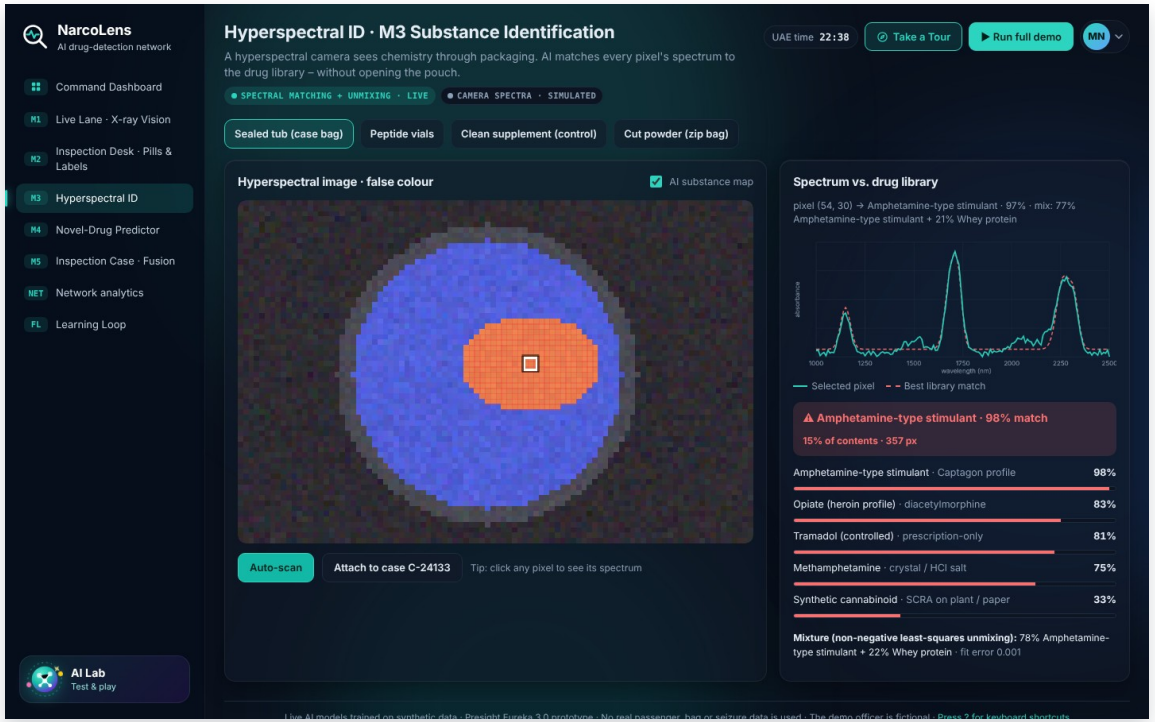
A convolutional segmentation network (28.7k parameters) marks where concealed organic material sits. On held-out synthetic data it flagged 311 of 311 concealment bags, with 1.7% false alarms on clean bags.



M2 · Tablet and label checks

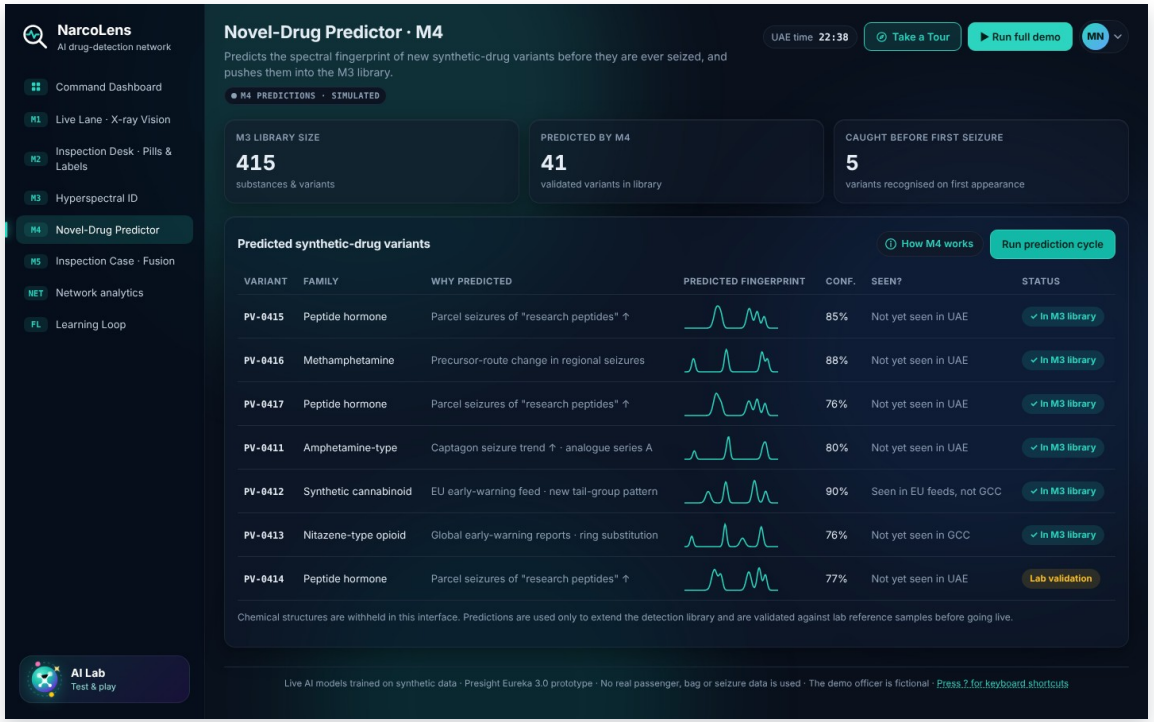
A convolutional classifier (88.1k parameters) verifies imprints, and a physical profile is measured from the millimetre scale. It reached 99.8% accuracy on 1,200 synthetic images. Label reading is simulated.

Seeing chemistry: identify, then anticipate



M3 · Hyperspectral identification

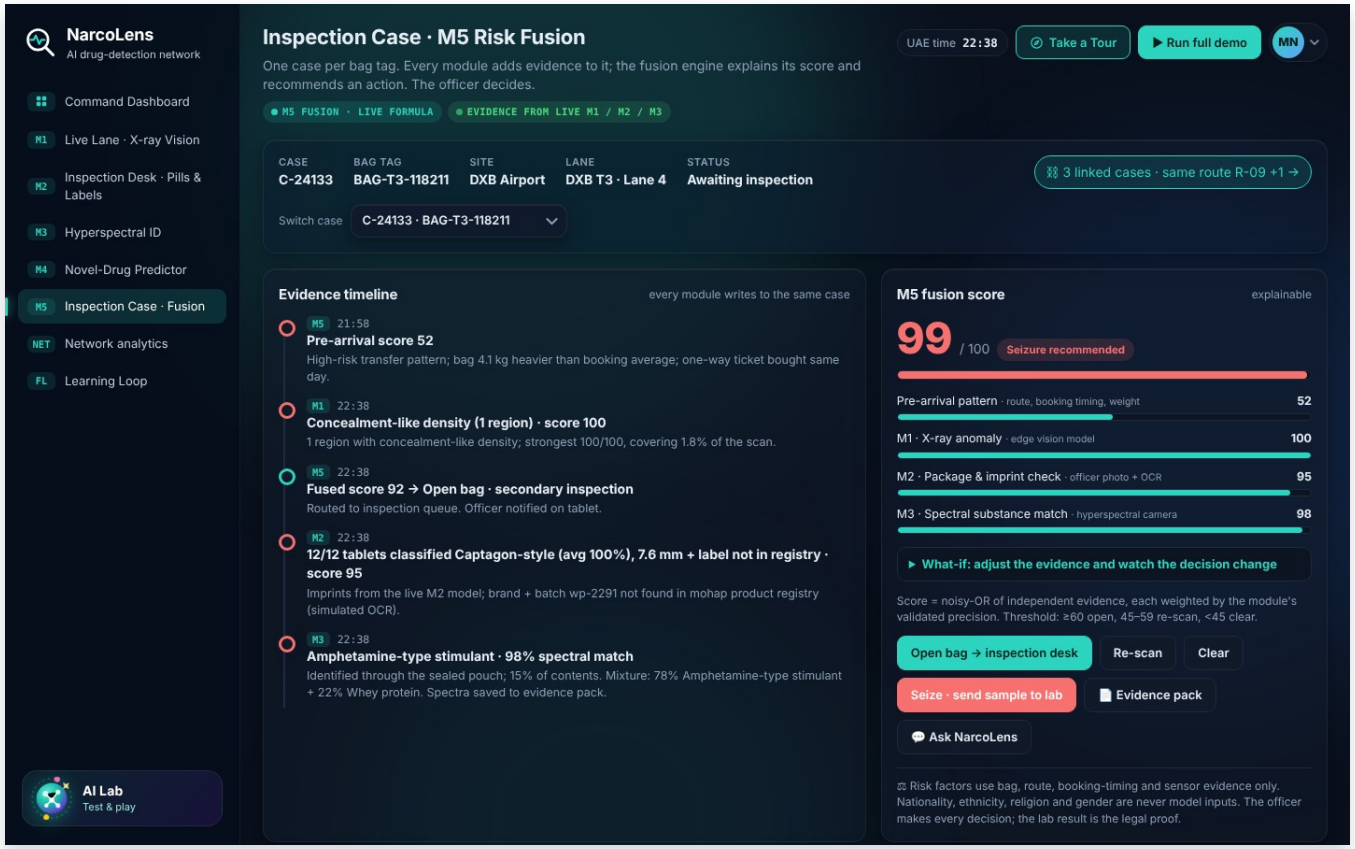
Each pixel's infrared spectrum is matched to a drug library by correlation, and mixtures are separated by non-negative least squares. The prototype library uses synthetic profiles.



M4 · Variant predictor

Proposes a graph neural network that predicts the spectral signature of emerging synthetic variants for review by forensic chemists. This cycle is simulated in the prototype.

Explainable by design, accountable by default



Explanation by default

A heatmap shows where the model looked, and the score shows each module's share.

The officer decides

NarcoLens flags, assesses and records. Seizures need supervisor approval.

Fair inputs

No nationality or ethnicity data is used.

Auditable record

Every action is logged in an evidence pack with a chain of custody.

Fusion model: $S = 1 - \prod (1 - w_i\, p_i)$, a weighted noisy-OR over the available evidence. The weights are design parameters and have not yet been calibrated on real outcomes.

Architecture: border sites feed a sovereign core

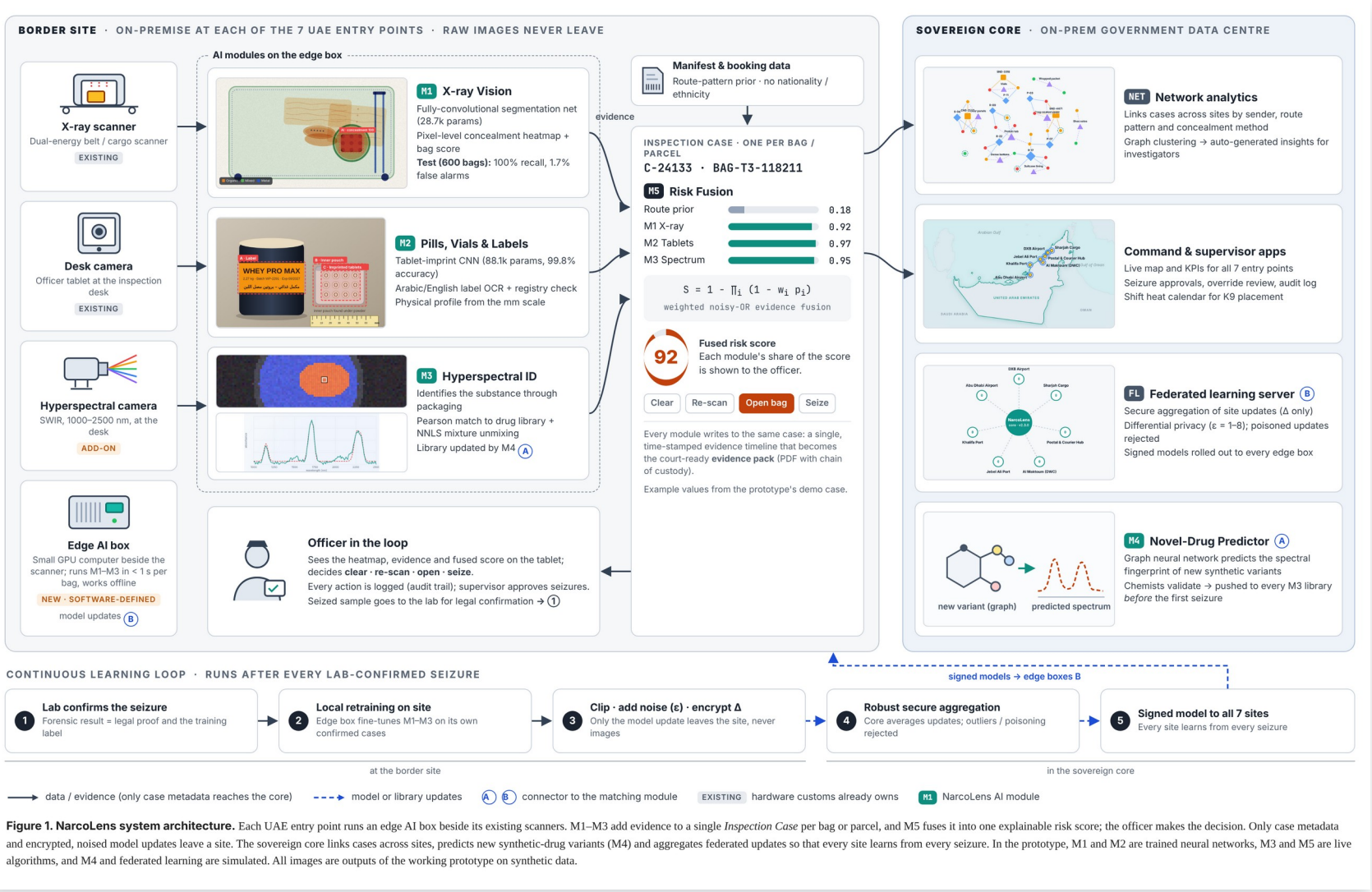


Figure 1. NarcoLens system architecture. Each UAE entry point runs an edge AI box beside its existing scanners. M1–M3 add evidence to a single *Inspection Case* per bag or parcel, and M5 fuses it into one explainable risk score; the officer makes the decision. Only case metadata and encrypted, noised model updates leave a site. The sovereign core links cases across sites, predicts new synthetic-drug variants (M4) and aggregates federated updates so that every site learns from every seizure. In the prototype, M1 and M2 are trained neural networks, M3 and M5 are live algorithms, and M4 and federated learning are simulated. All images are outputs of the working prototype on synthetic data.

At each site

An edge computer runs M1 to M3 beside the scanner. Raw images stay on site.

In the sovereign core

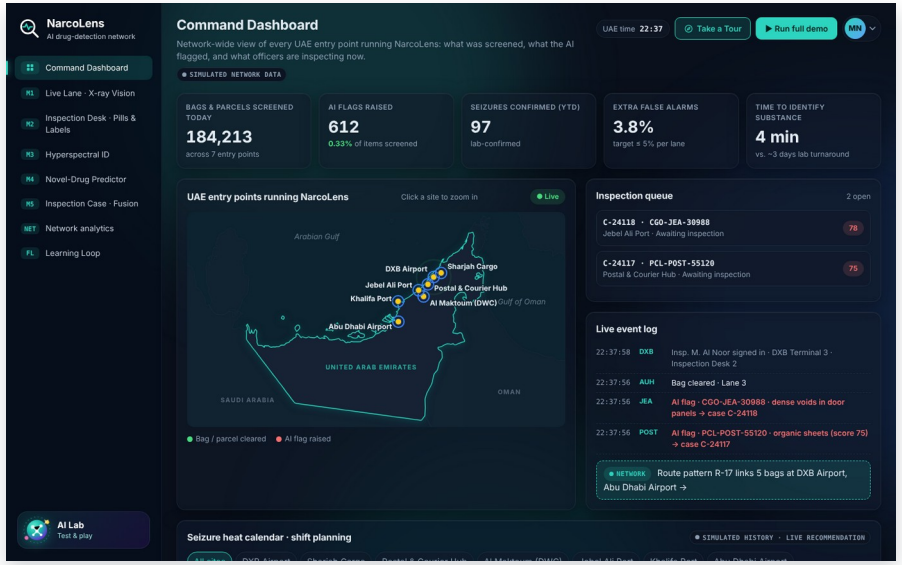
Network analytics, command and supervisor apps, federated aggregation and the variant predictor.

What leaves a site

Only case metadata and protected model updates.

Working prototype: what is live and what is simulated

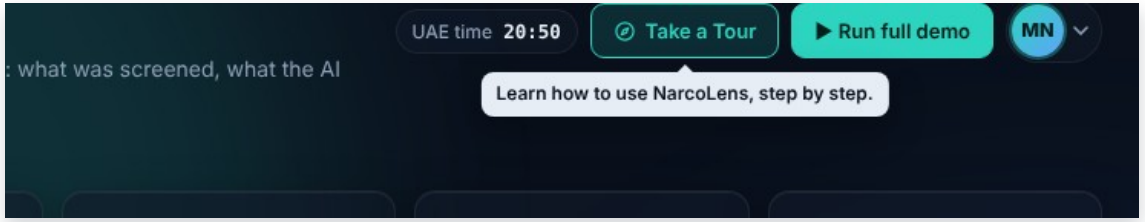
Part	Technology	Status
M1 X-ray vision	Segmentation network	Trained, live
M2 imprint check	Convolutional classifier	Trained, live
M2 label reading	Text recognition	Simulated
M3 hyperspectral	Spectral matching, unmixing	Live algorithm
M4 variant predictor	Graph neural network	Simulated
M5 risk fusion	Weighted noisy-OR	Live algorithm
Network analytics	Graph clustering	Live, simulated cases
Federated learning	Secure aggregation, privacy	Simulated



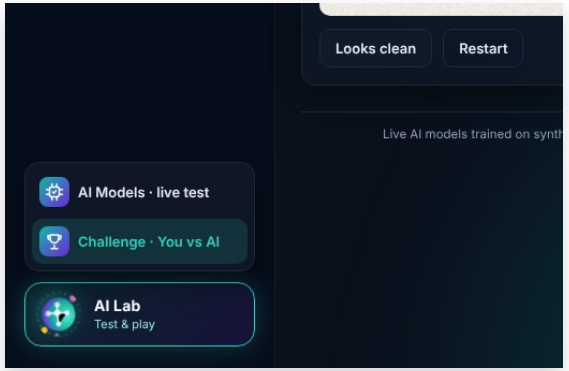
Command dashboard (simulated network data)

Runs offline in a browser. All data is synthetic or simulated, and the officer personas are fictional. Every screen states what is live and what is simulated.

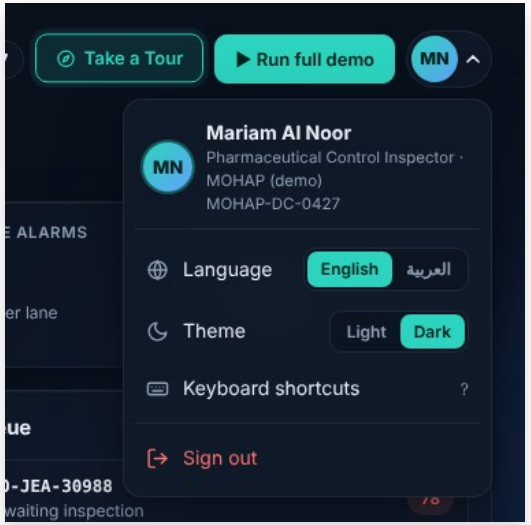
Try it in five minutes



Take a Tour: a five-step guided walk-through



AI Lab: live model test and a challenge against the AI



User menu: language (English or Arabic) and theme



Learning Loop: an interactive federated-learning simulator

- Press Run full demo to follow one bag end to end
- Build your own bag, hide drugs and scan it with the live model
- Draw a tablet imprint and see the classifier respond

Measured on synthetic data; real-data validation planned

Model	Test set	Metric	Result
M1 X-ray	600 held-out bags	Bag recall	**100% (311/311)**
		False alarms, clean bags	**1.7%**
		Pixel IoU	**0.916**
M2 imprint	1,200 tablet images	Four-class accuracy	**99.8%**

How to read these numbers

- Real concealment scans were not available, so the models learned from synthetic images with exact pixel-level labels.
- The test sets are synthetic too. They show that the method works. They do not estimate accuracy on real scans.

Path to validation

TRL 3 today

Proof of concept on synthetic data

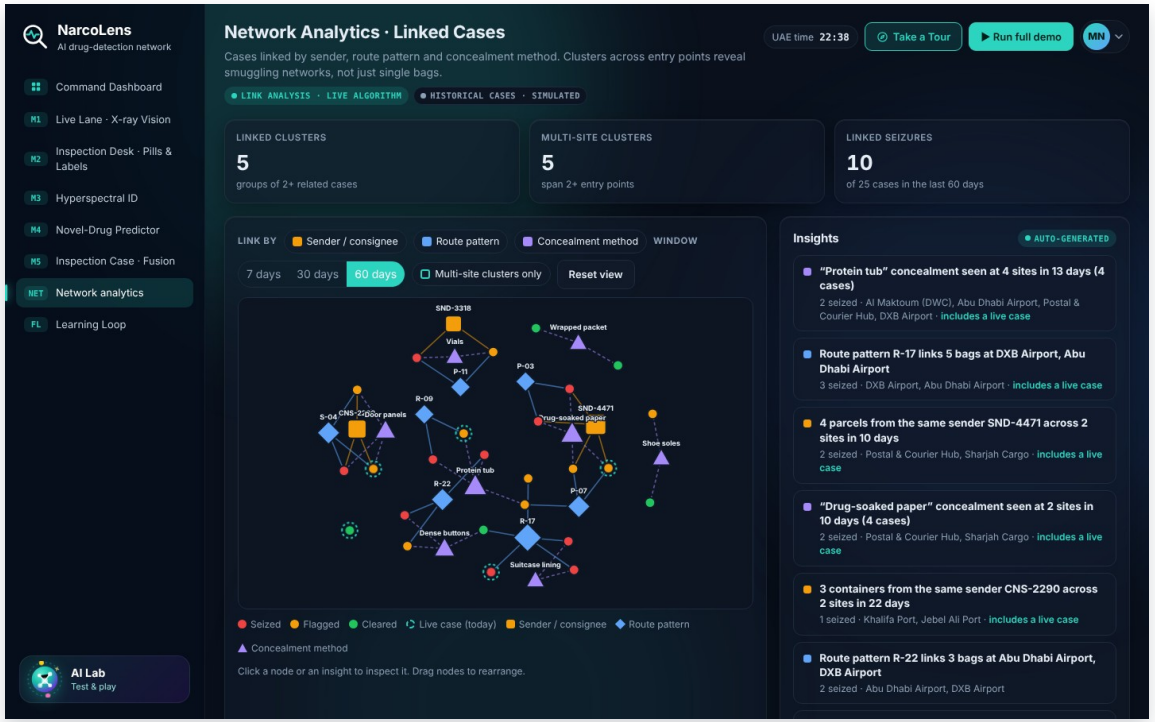
Shadow-mode trial

Runs beside current inspections on real scans without making decisions

TRL 4 and beyond

Recall at an agreed false-alarm rate, by scanner vendor; fusion weights calibrated on real outcomes

Linking cases and learning across sites





Network analytics

Cases are linked by shared sender, route pattern and concealment method. Clusters become insights for investigators, such as parcels from one sender across two sites.

Federated learning

Aims to let each site learn from laboratory-confirmed seizures. Only protected model updates leave a site, and a poisoned update can be rejected. A privacy slider trades learning speed for protection.

Intended impact, stated as hypotheses for the pilot

Fewer unnecessary searches

Evidence-led flags could reduce manual searches of legitimate travellers and parcels.

Faster assessment

Identifying a flagged item at the desk, before laboratory results, could shorten the time to a decision.

Stronger records

One case record could make the basis of each decision easier to review and refer.

Shared learning

Every confirmed seizure could improve detection at every site.

These are hypotheses to test in a shadow-mode pilot. They are not measured results. A proposed acceptance criterion is a false-alarm rate of no more than 5% per lane.

Who benefits

- Customs inspectors and X-ray operators
- Police anti-narcotics teams and supervisors
- Laboratory analysts and investigators
- Travellers, through fewer manual searches

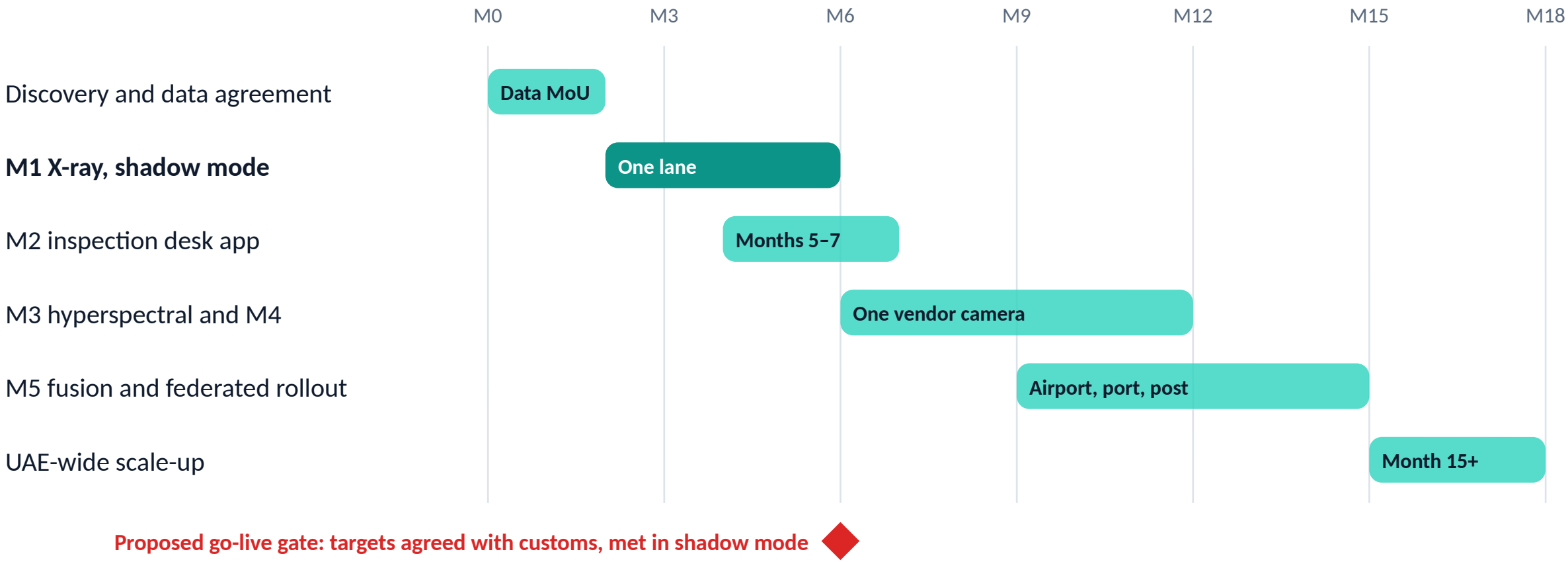
Value for Presight

- A sovereign product for UAE border security
- Network leads across entry points
- A path to GCC customs agencies

Beyond narcotics

- Weapons and explosives
- Counterfeit goods and wildlife trafficking
- Undeclared currency and gold

Roadmap: shadow-mode pilot to UAE-wide rollout



Team at peak of about 12: computer vision, spectroscopy ML, data and MLOps engineering, full-stack, a product lead with a customs expert, and a responsible-AI and legal lead

Risks we have designed for

Risk	How NarcoLens handles it
Little labelled real X-ray data	Pretraining, threat-image projection and active learning from officer feedback; real-data validation in the pilot
False alarms slow the queues	Shadow mode first; per-lane thresholds; officers see the highest-risk items
Closed scanner formats	The DICOS security-imaging standard; early partnership with one or two scanner vendors
Profiling and bias	No nationality or ethnicity inputs; regular bias audits; full audit log
Evidence in court	AI flags only; laboratory confirmation stays the legal proof; heatmaps and chain of custody

Why Presight

- Sovereign, on-premise AI for UAE government is core business
- Big-data analytics for public safety and investigations
- In-house talent in computer vision, Arabic NLP and MLOps

The team and the ask

Team

Farah Shaik

Riazat Rasool

Presight Eureka 3.0 · Defense & Policing

Our ask

1. **An ELT sponsor**
   to champion NarcoLens with UAE border agencies

2. **An introduction to customs**
   and border authorities, to agree a data-sharing arrangement for historical scans

3. **A shadow-mode pilot**
   on one lane, running beside current inspections with no effect on passengers

One confirmed seizure should make every UAE border smarter.